Scalable Machine Learning: Gradient Descent Optimization and Regularization in Deep Networks

1. Abstract & Introduction

Supervised and unsupervised Machine Learning models form the foundation of predictive analytics. Stochastic Gradient Descent (SGD), AdamW, and RMSprop optimization techniques are critical for training deep neural networks. In this work, we analyze convergence rates, loss landscape geometry, and hyperparameter tuning strategies across complex empirical datasets.

2. Loss Functions & Regularization

To prevent overfitting, we enforce L2 weight decay, dropout regularization (p=0.3), and batch normalization layers. Principal Component Analysis (PCA) and K-means clustering are utilized for exploratory feature reduction and unlabelled data partitioning.

3. Experimental Evaluation

Training convergence was measured across 100 epochs. AdamW optimizer achieved a cross-entropy loss of 0.12 with 94.8% test accuracy on multi-class benchmark datasets.

4. Conclusion

Proper regularization and adaptive learning rate schedulers drastically reduce generalization error in modern deep learning pipelines.